

2024

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : SEC-3P

(Mobile App Development)

Full Marks : 50

The figures in the margin indicate full marks.

SET - I

Marks Distribution :

User Interface (UI) / Front End	:	05
Procedure/Steps	:	10
Source Code	:	10
Output	:	07
Conclusion/Discussion	:	03
Experiment Total	:	35
Laboratory Notebook	:	05
Viva voce	:	10
Total	:	50

Answer *any one* question.

1. Develop a Flutter app using Dart programming that allows users to convert temperatures between Celsius and Fahrenheit. The app should include an input field for the temperature value, drop-down menus to select the input & output and a Submit button to perform the conversion. 35
2. Develop a Flutter app using Dart programming to build a basic mathematical calculator app that can perform addition, subtraction, multiplication, and division. Use 'Text Field' for input and 'Raised Button' for operations. 35

3. Develop a Flutter app using Dart programming to calculate the Body Mass Index (BMI) based on user input for weight and height. The app should include input fields for users to enter their weight (in kilograms) and height (in cm), a button to trigger the BMI calculation and a display area for the BMI result along with its classification (e.g. Underweight, Normal weight, Overweight, Obese). If BMI < 18.5 show underweight, BMI 18.5 ≤ 24.9 show normal weight, and BMI 24.9 ≤ 29.9 show overweight, else obese. 35
 4. Develop a Flutter app using Dart programming to create a form with multiple input fields (name, Age) and calculate he or she is eligible for voter or not.
(If Age ≥ 18 then display "Eligible for vote" otherwise show "not Eligible for vote") 35
 5. Develop a Flutter app using Dart programming that has a button and a text widget. When the button is pressed, the text widget should display the number of times the button has been pressed. 35
 6. Create a Flutter app using Dart programming that calculates a person's age based on his/her date of birth. The app should include an input field, a button to trigger the calculation and a display area to show the result. 35
 7. Develop a Flutter app using Dart programming that functions as a stopwatch with start and stop function. The app should include a display area to show start and stop button and an area where the time duration between start and stop will be shown. 35
 8. Develop a Flutter app using Dart programming that converts Indian currency (INR) to US Currency (USD) based on user defined exchange rates (87 INR = 1 USD). The app should include input fields for users to enter the amount, drop-down menus to select the input and output currencies, and a button to perform the conversion. Ensure that the app displays the converted amount. 35
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